

Multifaceted Exploration of Spatial Openness in Rental Housing: A Big Data Analysis in Tokyo’s 23 Wards

賃貸住宅における空間開放性の多面的探究：東京23区のビッグデータ分析

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1 Introduction

Understanding spatial openness in residential housing is important for enhancing living quality and guiding architectural design. Traditional indicators such as floor area and room count fail to capture the nuanced spatial qualities present in interior imagery and floor plans. While existing research has explored aspects of spatial openness, a comprehensive evaluation framework is lacking. This study proposes a multifaceted evaluation method of spatial openness in rental housing by leveraging large-scale data from Tokyo’s 23 wards.

The study constructs an analytical method that quantitatively extracts housing openness from both 2D and 3D information. Based on a large-scale rental housing dataset from Tokyo’s 23 wards, it investigates how openness features vary across geographic regions and construction decades, identifying spatial and temporal patterns. Furthermore, it explores correlations between spatial openness, rental cost, and subjective impression scores. In conclusion, this study proposes a comprehensive framework for evaluating spatial openness, making full use of large-scale rental housing data from Tokyo’s 23 wards.

2 Methodology

2.1 Dataset and Attributes

Our dataset comprises rental properties from Tokyo’s 23 wards (1960-present), filtered for image quality and data completeness, as shown in Figure ??.

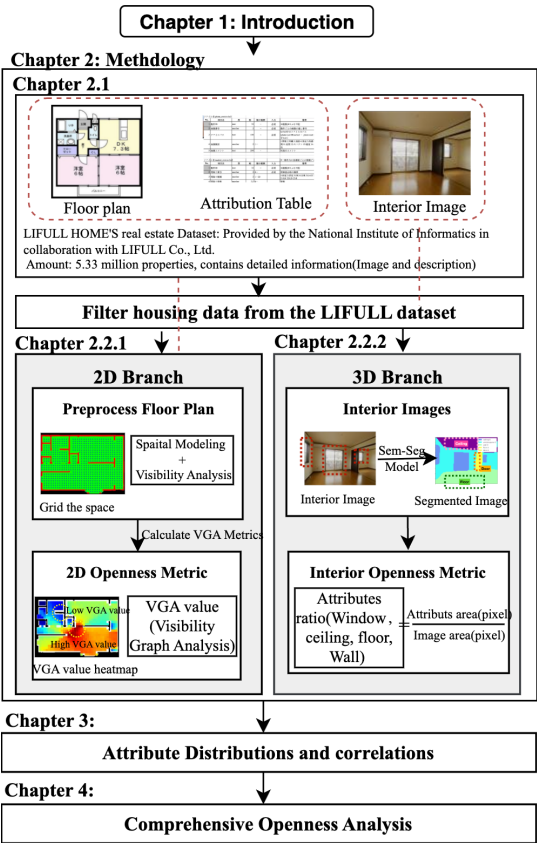


Figure 1: Data selection and filtering process.

As illustrated in Figure ??, this overview shows our methodology extracts multiple categories of spatial openness attributes through 3D interior images, 2D floorplan data, and property specifications, as well as our following data analysis.

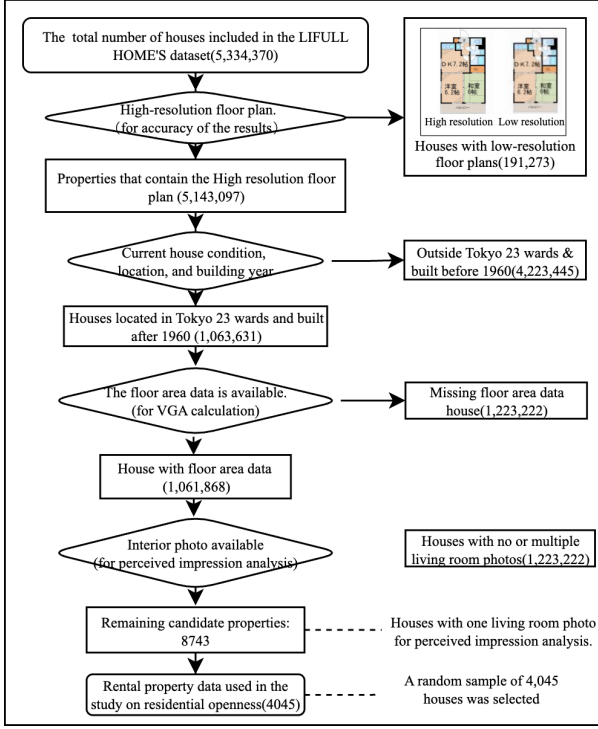


Figure 2: Feature extraction overview.

Impression scores (Q3) are extracted using Shimomura et al.’s pre-trained model. The following attributes are the focus of our analysis: **room size attributes, property specifications, interior elements, and vga-based statistics**. Room size attributes and property specifications are obtained directly from the database. Interior elements are extracted through semantic segmentation of interior images, while VGA-based statistics are computed from floorplan analysis to quantify spatial connectivity.

2.2 Interior Elements

We apply Mask2Former semantic segmentation to extract wall, ceiling, floor, and window ratios from interior images. The fractions of each visual element are calculated by the fraction of pixels from the segmentation mask, as shown in Figure ??.

2.3 VGA-based Statistics

For floorplan data, we use semantic segmentation to identify walls, rooms, and open areas. We then create a grid over the open areas and calculate VGA (Visibility Graph Analysis) values:

$$S(i) = \sum_{j=1}^N V_{ij} \quad (1)$$

where $S(i)$ is the visibility score at node i , and $V_{ij} = 1$ if node j is visible from node i , otherwise $V_{ij} = 0$. This creates a VGA heatmap showing spatial connec-

tivity throughout the floorplan. We extract summary statistics (mean, standard deviation, percentiles) from this heatmap as spatial openness features, as shown in Figure ??. After extracting the statistics, we can utilize them to analyze their distributions and correlations with other attributes to enhance the understanding of spatial openness. of the openness of the properties.

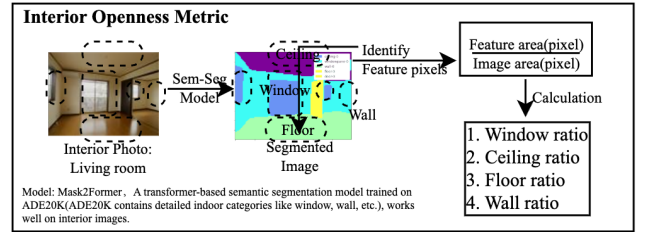


Figure 3: Interior semantic segmentation and impression scoring.

3 Attribute Distributions and Correlations

3.1 Temporal Analysis of Spatial Openness

From the tabular features of the properties, we have various information, with the most interesting being building age and geographic factors. We examined the distribution of VGA mean values across construction dates in decades, as shown in Figure ??.

The analysis shows VGA mean values contain large outliers but remain stable across decades, likely due to large-area properties. The impression scores show more balanced distributions, as they are evaluated from interior images that are less dependent of overall property layouts.

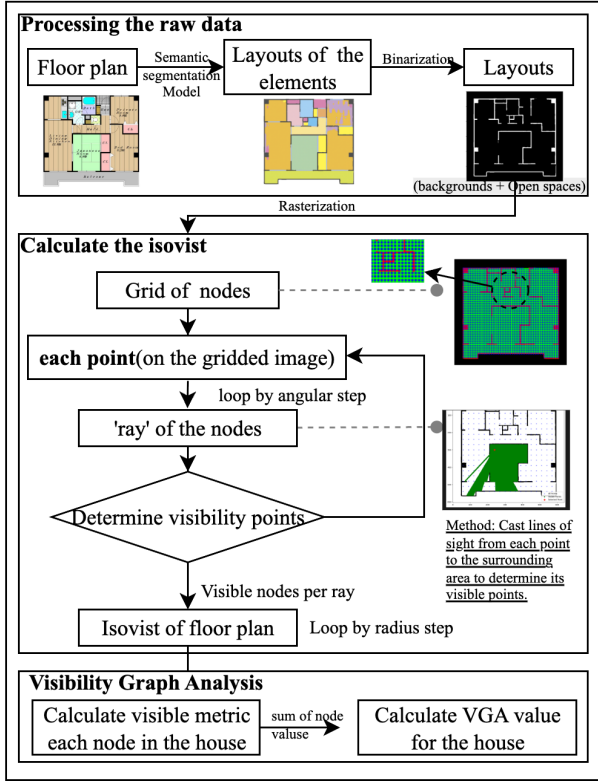


Figure 4: Workflow of creating VGA heatmap from floorplan data: raw floorplan, semantic segmentation output, physical gridding, and VGA heatmap.

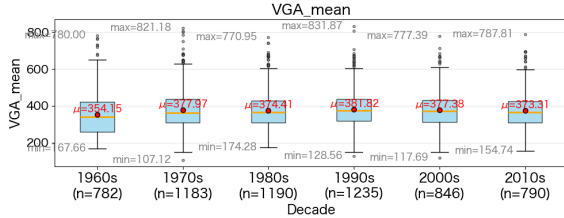


Figure 5: Distribution of VGA mean values across construction dates by decade

3.2 Geographic Distribution Analysis

We conducted a comprehensive analysis of the geographic distribution of VGA mean values and impression scores at the chome-level granularity using post-code alignment across Tokyo's 23 special wards, as shown in Figure ???. This fine-grained spatial analysis allows us to examine potential neighborhood-level variations in spatial openness characteristics. However, despite this detailed geographic examination, no clear clustering patterns or significant imbalances are found in the distributions across the different areas for either attribute, suggesting that spatial openness characteristics are relatively uniformly distributed across Tokyo's urban landscape.

3.3 Correlation Analysis Between Openness Attributes

We conducted a detailed correlation analysis examining the relationships between VGA statistics, impression scores, and interior element fractions to understand how different measures of spatial openness relate to each other. Figure ?? reveals only mild correlations between VGA metrics derived from floorplan analysis and impression scores obtained from interior imagery assessment, suggesting fundamental gaps between objective floorplan-based spatial analysis and subjective imagery-based perceptual assessment of spatial openness.

4 Comprehensive Openness Analysis

4.1 Analysis of Impression Scores and Interior Elements

We performed core-component dependency analysis combining impression scores with interior element ratios after applying rigorous outlier filtering using a 75th percentile cutoff to ensure data quality. The cluster analysis reveals distinct groupings based on floor and ceiling visibility characteristics, with moderate negative correlations observed with impression scores, indicating that higher visibility of structural elements tends to correspond with lower perceived spatial openness, as demonstrated in Figure ?? and quantified in Table ??.

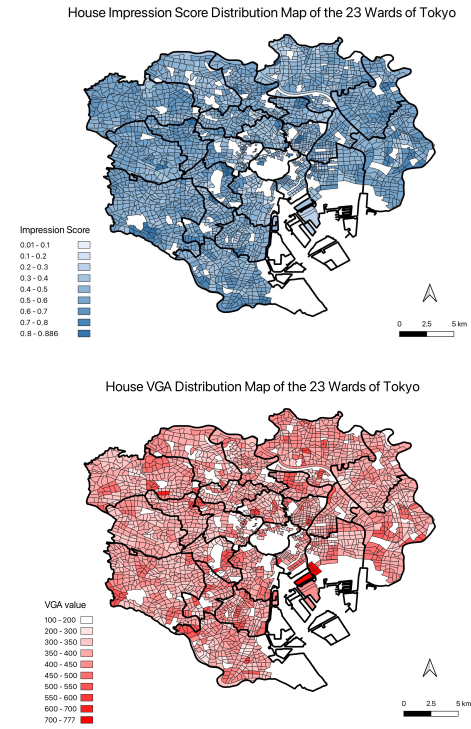


Figure 6: Geographic distribution analysis across Tokyo's 23 special wards: impression scores (top) and VGA mean values (bottom).

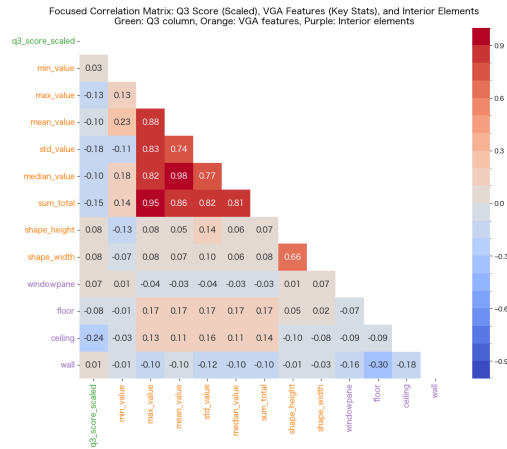


Figure 7: Correlation matrix between Q3 impression scores, VGA metrics (mean, std, min, max), and interior element fractions

Interior Element	Correlation	Sample Size (n)
Windowpane	0.0905	3012
Floor	-0.1577	4439
Ceiling	-0.3043	3559
Wall	0.0275	4430

Table 1: Correlations between interior element ratios and Q3 impression scores after 75% filtering

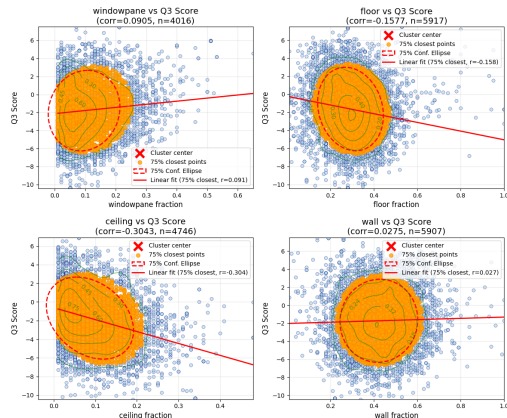


Figure 8: Cluster analysis of impression scores vs. interior element ratios

4.2 Regression Analysis Against Rent Cost

We performed regression analysis to predict rent cost using openness attributes. XGBoost achieved the best fit with acceptable R^2 scores. Analysis reveals that standard deviation and total sum of VGA values are the most important factors for rent prediction, thus there is a possibility that larger rooms with

more spatial subdivisions tend to increase rental prices in a human understandable way. The distance towards the top 2 stations falls behind the above factors slightly. Additionally, our analysis reveals that user impression scores contribute to rent price trends moderately, suggesting a meaningful relationship between subjective spatial perception and market valuation. This indicates that user-perceived spatial quality aligns with economic value assessment in the rental market. As shown in Figure ??, training and testing results demonstrate reasonable performance. Residual analysis reveals weak bias in the fit showing that the model is not overfitting.

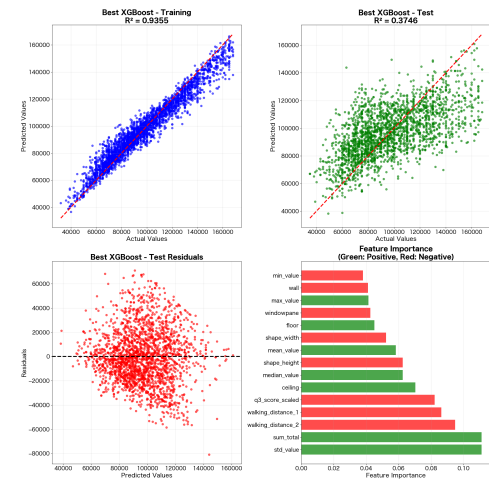


Figure 9: XGBoost regression results for rent prediction using openness attributes

4.3 Principal Component Analysis

To better understand the relationships between spatial features, we performed Principal Component Analysis (PCA) using VGA mean values, ceiling, wall, floor, and window visual fractions as input features. The analysis reveals the underlying structure of spatial characteristics in our dataset. The PCA plots are shown as Figure ??.

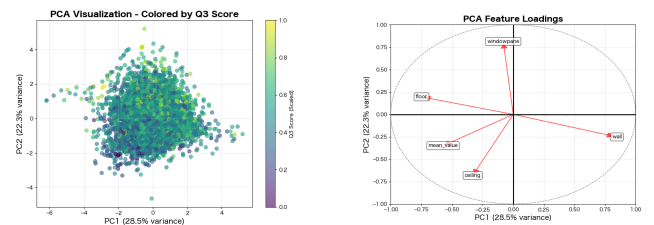


Figure 10: PCA distribution and top 2 components with raw feature axis

Table ?? presents the explained variance for each principal component. The first two components capture approximately 51% of the total variance.

Comp.	Eigenvalue	Var. %	Cum. %
PC1	1.42	28.46	28.46
PC2	1.12	22.33	50.80
PC3	1.07	21.36	72.16
PC4	0.88	17.57	89.73
PC5	0.51	10.27	100.00

Table 2: Principal component variance explanation

Table ?? shows the feature loadings for all principal components. PC1 is primarily dominated by wall fractions (0.65) opposing floor visibility (-0.57). PC2 captures the balance between natural lighting (window: 0.70) and ceiling prominence (-0.59).

Name	PC1	PC2	PC3	PC4	PC5
VGA. Avg	-0.44	-0.30	-0.15	0.82	-0.15
Window	-0.06	0.70	0.50	0.37	0.35
Floor	-0.57	0.17	-0.53	-0.23	0.56
Ceiling	-0.25	-0.59	0.62	-0.16	0.43
Wall	0.65	-0.22	-0.27	0.32	0.60

Table 3: Principal component feature loadings

test table

Feature	VIF	Status
min_all	1.35	Low
max_all	28.16	High
mean_all	8.34	Moderate
std_all	5.58	Moderate
min_lv	3.40	Low
max_lv	37.94	High
std_lv	3.63	Low
median_lv	18.29	High
window	1.06	Low
floor	1.21	Low
ceiling	1.13	Low
wall	1.23	Low

Table 4: Variance Inflation Factor (VIF) for multicollinearity assessment, VIF > 10: High, VIF > 5: Moderate, VIF < 5: Low

Feature	VIF	Status
min_value_raw_all	1.33	Low
max_value_raw_all	9.24	Moderate
mean_value_raw_all	5.42	Moderate
std_value_raw_all	4.09	Low
min_value_raw_lv	2.86	Low
std_value_raw_lv	2.92	Low
ceiling	1.03	Low

Table 5: Variance Inflation Factor (VIF) for multicollinearity assessment (after feature selection), VIF > 10: High, VIF > 5: Moderate, VIF < 5: Low

Feature	Coef. ($\times 10^3$)	Std ($\times 10^2$)	t-stat	p-value	Sig.
min_all	-2.68	5.90	-4.55	0.00	Y
max_all	26.70	27.00	9.89	0.00	Y
mean_all	-23.80	14.70	-16.19	0.00	Y
std_all	6.77	12.00	5.64	0.00	Y
min_lv	-6.20	9.38	-6.61	0.00	Y
max_lv	-4.08	31.30	-1.30	0.19	N
std_lv	-7.97	9.69	-8.23	0.00	Y
median_lv	18.10	21.70	8.31	0.00	Y
window	-0.07	5.24	-0.14	0.89	N
floor	-2.01	5.59	-3.59	0.00	Y
ceiling	7.49	5.41	13.83	0.00	Y
wall	-0.10	5.63	-0.19	0.85	N

Table 6: Feature coefficients for rent prediction (normalized features), their t-stats and p-values in the test against the impact towards the rent price. Confidence level set to 95%.

Feature	Coef. ($\times 10^3$)	Std ($\times 10^2$)	t-stat	p-value	Sig.
min_all	-2.15	5.95	-3.61	0.00	Y
max_all	25.48	15.67	16.27	0.00	Y
mean_all	-16.63	12.00	-13.86	0.00	Y
std_all	12.13	10.42	11.64	0.00	Y
min_lv	-3.37	8.72	-3.86	0.00	Y
std_lv	-6.80	8.81	-7.72	0.00	Y
ceiling	8.30	5.23	15.86	0.00	Y

Table 7: Feature coefficients (after feature selection) for rent_plus_management_fee prediction (normalized features), their t-stats and p-values in the test against the impact towards the rent price. Confidence level set to 95%.

Abbr.	Description
min_all	Min VGA (whole property)
max_all	Max VGA (whole property)
mean_all	Mean VGA (whole property)
std_all	Std VGA (whole property)
min_lv	Min VGA (living room)
max_lv	Max VGA (living room)
std_lv	Std VGA (living room)
median_lv	Median VGA (living room)
window	Window fraction
floor	Floor fraction
ceiling	Ceiling fraction
wall	Wall fraction

Table 8: Feature abbreviations and descriptions.

5 Conclusions

This research presents a framework for analyzing spatial openness in rental housing through automated

madori interpretation, visibility graph analysis, and interior semantic segmentation. Our methodology quantifies subjective spatial qualities using visual data from rental platforms. Key findings reveal weak correlations between VGA metrics and user impression scores, with excessive floor/ceiling visibility negatively impacting quality perceptions. PCA analysis reveals that the first two components capture 51% of variance, with PC1 dominated by wall-floor opposition and PC2 capturing natural lighting versus ceiling prominence. Regression analysis demonstrates that VGA standard deviation and room size are the most important factors for rent prediction. Spatial openness perception is impacted by the complex interactions between these factors. Our results identify their distinct contributions, revealing how visual effects and VGA metrics shape openness evaluation. This research provides practical tools for data-driven spatial openness analysis, which is a step towards openness-oriented housing design.